AGL_anomaly_detection

April 13, 2026

1 Adaptive Guardrail Layer (AGL) - Anomaly Detection Notebook

1.1 Purpose

This notebook develops and compares unsupervised anomaly detection models for malicious prompt detection using the engineered feature dataset. It is kept compatible with Google Colab and assumes the input dataset is available at `/content/dataset_feature_engineered.csv`.

1.2 Goals

- Load the engineered dataset
- Build a benign-only training set for anomaly detection
- Compare various models
- Tune thresholds on validation data
- Evaluate final model behavior on held-out test data

1.3 Expected input

- `dataset_feature_engineered.csv`

[2]:
```
import os
import random
import warnings
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display
from sklearn.model_selection import train_test_split, ParameterGrid
from sklearn.preprocessing import RobustScaler, StandardScaler,␣
 ↪QuantileTransformer
from sklearn.ensemble import IsolationForest
from sklearn.neighbors import LocalOutlierFactor
from sklearn.metrics import (
    classification_report,
    confusion_matrix,
    precision_recall_curve,
```

```
    f1_score,
    precision_score,
    recall_score,
    roc_auc_score,
    average_precision_score,
)
from sklearn.feature_selection import VarianceThreshold
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, TensorDataset

warnings.filterwarnings("ignore")

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)

VAL_SIZE = 0.20
TEST_SIZE = 0.20

# runtime controls
MAX_BENIGN_TRAIN_SAMPLES_IF = None
MAX_BENIGN_TRAIN_SAMPLES_LOF = 80000
BATCH_SIZE = 256

# autoencoder controls
AE_EPOCHS = 120
AE_PATIENCE = 12
AE_LEARNING_RATE = 8e-4
AE_WEIGHT_DECAY = 1e-5
AE_NOISE_STD = 0.03

THRESHOLD_OBJECTIVE = "f1"   # f1 or recall_priority
MIN_RECALL = 0.70
```

1.3.1 Load data

[4]:
```
INPUT_PATH = "../../data/processed/"
INPUT_FILENAME = Path(INPUT_PATH) / "dataset_feature_engineered.csv"

if not os.path.exists(INPUT_FILENAME):
    raise FileNotFoundError(f"Could not find input file: {INPUT_FILENAME}")
```

```
df = pd.read_csv(INPUT_FILENAME)
print(f"Loaded dataset: {df.shape[0]:,} rows x {df.shape[1]:,} columns")
display(df.head())
```

```
Loaded dataset: 104,755 rows x 74 columns
```

```
   row_id                                             prompt  label  \
0       0  Here's what I need you to do for me: 1. Transl…      0
1       1  I need some information quickly: 1. What is th…      0
2       2  Please do the following tasks: 1. Explain what…      0
3       3  Could you help with these items? 1. Define 'Ma…      0
4       4  I'm curious about a few things: 1. What's the …      0

  source_dataset  char_length  word_count  avg_word_length  line_count  \
0   WildGuardMix          220          45         3.911111           1
1   WildGuardMix          188          36         4.250000           1
2   WildGuardMix          159          29         4.517241           1
3   WildGuardMix          185          35         4.314286           1
4   WildGuardMix          156          31         4.064516           1

   sentence_count  token_count  …  embed_pca_23  embed_pca_24  embed_pca_25  \
0               8           59  …     -0.024124      0.085803     -0.219357
1               8           48  …      0.120953     -0.182473     -0.120312
2               8           38  …      0.216458      0.016013      0.024323
3               9           50  …      0.111428      0.043126     -0.114772
4               8           48  …      0.126424     -0.039547     -0.020182

   embed_pca_26  embed_pca_27  embed_pca_28  embed_pca_29  embed_pca_30  \
0     -0.063032     -0.174054     -0.014205      0.049955      0.093883
1      0.031146     -0.101867      0.023967     -0.085226      0.004245
2     -0.083344     -0.111400      0.011361     -0.002220      0.093972
3     -0.082916     -0.109635      0.036963     -0.004525     -0.094766
4     -0.033340     -0.072246      0.006569     -0.042061      0.071661

   embed_pca_31  embed_pca_32
0      0.005645      0.003084
1     -0.079983      0.004235
2     -0.068973      0.044231
3     -0.133499     -0.075309
4      0.026669      0.036026

[5 rows x 74 columns]
```

1.3.2 Define columns for unsupervised training

[6]:
```
target_column = "label"

feature_columns = [
    "token_count",
    "token_word_ratio",
    "whitespace_ratio",
    "punctuation_ratio",
    "special_char_ratio",
    "non_ascii_ratio",
    "markdown_symbol_count",
    "has_code_block",
    "repeated_punct_count",
    "quote_count",
    "bracket_count",
    "colon_count",
    "semicolon_count",
    "instruction_override_score",
    "roleplay_score",
    "payload_request_score",
    "social_engineering_score",
    "obfuscation_score",
    "has_instruction_override",
    "has_roleplay",
    "has_payload_request",
    "has_social_engineering",
    "has_obfuscation",
    "embedding_norm",
    "embed_pca_1",
    "embed_pca_2",
    "embed_pca_3",
    "embed_pca_4",
    "embed_pca_5",
    "embed_pca_6",
    "embed_pca_7",
    "embed_pca_8",
    "embed_pca_9",
    "embed_pca_10",
    "embed_pca_11",
    "embed_pca_12",
    "embed_pca_13",
    "embed_pca_14",
    "embed_pca_15",
    "embed_pca_16",
    "embed_pca_17",
    "embed_pca_18",
```

```
    "embed_pca_19",
    "embed_pca_20",
    "embed_pca_21",
    "embed_pca_22",
    "embed_pca_23",
    "embed_pca_24",
    "embed_pca_25",
    "embed_pca_26",
    "embed_pca_27",
    "embed_pca_28",
    "embed_pca_29",
    "embed_pca_30",
    "embed_pca_31",
    "embed_pca_32",
]
print(f"Total initial features: {len(feature_columns)}")
```

```
Total initial features: 56
```

1.3.3 Train / validation / test split

- split on the full labeled dataset with stratification
- train unsupervised models (only on benign rows)
- keep validation/test untouched for threshold tuning and final evaluation

[8]:
```
train_df, temp_df = train_test_split(
    df,
    test_size=VAL_SIZE + TEST_SIZE,
    stratify=df[target_column],
    random_state=SEED,
)

relative_test_size = TEST_SIZE / (VAL_SIZE + TEST_SIZE)

val_df, test_df = train_test_split(
    temp_df,
    test_size=relative_test_size,
    stratify=temp_df[target_column],
    random_state=SEED,
)

train_benign_df = train_df.loc[train_df[target_column] == 0].copy()

print(f"Train rows (full): {len(train_df):,}")
print(f"Train benign rows: {len(train_benign_df):,}")
print(f"Validation rows: {len(val_df):,}")
print(f"Test rows: {len(test_df):,}")
```

```
print("\nValidation label distribution:")
print(val_df[target_column].value_counts(normalize=True).sort_index())

print("\nTest label distribution:")
print(test_df[target_column].value_counts(normalize=True).sort_index())
```

```
Train rows (full): 62,853
Train benign rows: 33,065
Validation rows: 20,951
Test rows: 20,951

Validation label distribution:
label
0    0.526037
1    0.473963
Name: proportion, dtype: float64

Test label distribution:
label
0    0.526085
1    0.473915
Name: proportion, dtype: float64
```

1.3.4 Shared helper functions

[11]:
```
def evaluate_model(y_true, y_pred, model_name, split_name):
    f1 = f1_score(y_true, y_pred, zero_division=0)
    prec = precision_score(y_true, y_pred, zero_division=0)
    rec = recall_score(y_true, y_pred, zero_division=0)

    print(f"\n--- {model_name} ({split_name}) ---")
    print(classification_report(y_true, y_pred, zero_division=0))

    return {
        "model": model_name,
        "split": split_name,
        "f1": f1,
        "precision": prec,
        "recall": rec,
    }

def plot_score_distributions(y_true, scores, title):
    plt.figure(figsize=(10, 6))
    sns.histplot(scores[y_true == 0], label="Benign", alpha=0.5,␣
 ↪stat="density", kde=True)
    sns.histplot(scores[y_true == 1], label="Malicious", alpha=0.5,␣
 ↪stat="density", kde=True)
```

```
    plt.title(f"Score Distribution: {title}")
    plt.xlabel("Anomaly score")
    plt.legend()
    plt.show()

def plot_confusion(y_true, y_pred, title):
    cm = confusion_matrix(y_true, y_pred)
    plt.figure(figsize=(5, 4))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
    plt.title(title)
    plt.xlabel("Predicted")
    plt.ylabel("Actual")
    plt.show()

def summarize_business_view(y_true, y_pred, positive_label_name="malicious"):
    tp = int(((y_true == 1) & (y_pred == 1)).sum())
    fn = int(((y_true == 1) & (y_pred == 0)).sum())
    fp = int(((y_true == 0) & (y_pred == 1)).sum())
    tn = int(((y_true == 0) & (y_pred == 0)).sum())

    rec = recall_score(y_true, y_pred, zero_division=0)
    prec = precision_score(y_true, y_pred, zero_division=0)

    print(f"Out of 100 {positive_label_name} prompts, the model catches about␣
 ↪{round(rec * 100)}.")
    print(f"Out of 100 prompts flagged as {positive_label_name}, about␣
 ↪{round(prec * 100)} are actually {positive_label_name}.")
    print(f"Confusion counts -> TP: {tp:,}, FN: {fn:,}, FP: {fp:,}, TN: {tn:,}")

def tune_threshold(y_true, scores, objective="f1", min_recall=0.60):
    precision, recall, thresholds = precision_recall_curve(y_true, scores)

    precision = precision[:-1]
    recall = recall[:-1]
    thresholds = np.array(thresholds)

    if len(thresholds) == 0:
        fallback = np.percentile(scores[y_true == 0], 95)
        preds = (scores > fallback).astype(int)
        return {
            "threshold": float(fallback),
            "precision": precision_score(y_true, preds, zero_division=0),
            "recall": recall_score(y_true, preds, zero_division=0),
            "f1": f1_score(y_true, preds, zero_division=0),
        }

    f1_values = (2 * precision * recall) / (precision + recall + 1e-12)
```

```
    if objective == "recall_priority":
        valid = np.where(recall >= min_recall)[0]
        if len(valid) > 0:
            best_idx = valid[np.argmax(f1_values[valid])]
        else:
            best_idx = int(np.argmax(f1_values))
    else:
        best_idx = int(np.argmax(f1_values))

    best_threshold = float(thresholds[best_idx])
    best_preds = (scores > best_threshold).astype(int)

    return {
        "threshold": best_threshold,
        "precision": precision_score(y_true, best_preds, zero_division=0),
        "recall": recall_score(y_true, best_preds, zero_division=0),
        "f1": f1_score(y_true, best_preds, zero_division=0),
    }

def score_summary(y_true, scores):
    return {
        "roc_auc": roc_auc_score(y_true, scores),
        "avg_precision": average_precision_score(y_true, scores),
    }

def fit_scaler(name, X_train, X_val, X_test):
    if name == "robust":
        scaler = RobustScaler()
    elif name == "standard":
        scaler = StandardScaler()
    elif name == "quantile":
        scaler = QuantileTransformer(
            output_distribution="normal",
            n_quantiles=min(1000, max(100, len(X_train))),
            random_state=SEED,
        )
    else:
        raise ValueError(f"Unknown scaler: {name}")

    X_train_scaled = scaler.fit_transform(X_train)
    X_val_scaled = scaler.transform(X_val)
    X_test_scaled = scaler.transform(X_test)

    return scaler, X_train_scaled, X_val_scaled, X_test_scaled

def robust_zscore(x):
```

```
    med = np.median(x)
    mad = np.median(np.abs(x - med)) + 1e-9
    return 0.6745 * (x - med) / mad
```

1.3.5 Feature Selection and Scaling

[14]:
```
X_train_benign_selected = train_benign_df[feature_columns].copy()
X_val_selected = val_df[feature_columns].copy()
X_test_selected = test_df[feature_columns].copy()

y_val = val_df[target_column].astype(int).to_numpy()
y_test = test_df[target_column].astype(int).to_numpy()

scaler, X_train_scaled, X_val_scaled, X_test_scaled = fit_scaler(
    "quantile",
    X_train_benign_selected,
    X_val_selected,
    X_test_selected,
)

print(f"Train benign shape: {X_train_scaled.shape}")
print(f"Validation shape: {X_val_scaled.shape}")
print(f"Test shape: {X_test_scaled.shape}")
```

```
Train benign shape: (33065, 56)
Validation shape: (20951, 56)
Test shape: (20951, 56)
```

1.4 Unsupervised learning

This notebook explores four distinct models and selects the best-performing one: 1. Isolation Forest 2. Local Outlier Factor (novelty mode) 3. Denoising Deep Autoencoder

1.4.1 1. Isolation Forest

[17]:
```
print("Training and tuning Isolation Forest...")

X_train_if = X_train_scaled
if MAX_BENIGN_TRAIN_SAMPLES_IF is not None and len(X_train_if) >␣
 ↪MAX_BENIGN_TRAIN_SAMPLES_IF:
    X_train_if = pd.DataFrame(X_train_if).sample(
        n=MAX_BENIGN_TRAIN_SAMPLES_IF,
        random_state=SEED,
    ).to_numpy()

if_params = {
    "n_estimators": [300, 500],
    "max_samples": ["auto", 0.8],
```

```
    "max_features": [0.8, 1.0],
    "bootstrap": [False, True],
    "contamination": ["auto", 0.01, 0.03],
}

best_if = None
best_if_model = None
best_if_scores_val = None

param_list = list(ParameterGrid(if_params))
total_runs = len(param_list)

for i, params in enumerate(param_list, 1):
    print(f"[{i}/{total_runs}] Isolation Forest params: {params}")

    candidate = IsolationForest(
        random_state=SEED,
        n_jobs=-1,
        **params,
    )
    candidate.fit(X_train_if)

    val_scores = -candidate.decision_function(X_val_scaled)
    threshold_info = tune_threshold(
        y_val,
        val_scores,
        objective=THRESHOLD_OBJECTIVE,
        min_recall=MIN_RECALL,
    )

    row = {
        "model": "Isolation Forest",
        **params,
        **threshold_info,
        **score_summary(y_val, val_scores),
    }

    print(
        f"   -> F1: {row['f1']:.4f} | Precision: {row['precision']:.4f} | "
        f"Recall: {row['recall']:.4f} | AP: {row['avg_precision']:.4f}"
    )

    if best_if is None or (row["f1"], row["avg_precision"]) > (best_if["f1"],␣
↪best_if["avg_precision"]):
        best_if = row
        best_if_model = candidate
        best_if_scores_val = val_scores
```

```
        print(f"***New BEST Isolation Forest (F1: {row['f1']:.4f})")

display(pd.DataFrame([best_if]))
```

```
Training and tuning Isolation Forest…
[1/48] Isolation Forest params: {'bootstrap': False, 'contamination': 'auto',
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6779 | Precision: 0.5672 | Recall: 0.8423 | AP: 0.6709
     New BEST Isolation Forest (F1: 0.6779)
[2/48] Isolation Forest params: {'bootstrap': False, 'contamination': 'auto',
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6820 | Precision: 0.5745 | Recall: 0.8388 | AP: 0.6798
     New BEST Isolation Forest (F1: 0.6820)
[3/48] Isolation Forest params: {'bootstrap': False, 'contamination': 'auto',
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6937 | Precision: 0.5821 | Recall: 0.8584 | AP: 0.7228
     New BEST Isolation Forest (F1: 0.6937)
[4/48] Isolation Forest params: {'bootstrap': False, 'contamination': 'auto',
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6959 | Precision: 0.6066 | Recall: 0.8161 | AP: 0.7315
     New BEST Isolation Forest (F1: 0.6959)
[5/48] Isolation Forest params: {'bootstrap': False, 'contamination': 'auto',
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6886 | Precision: 0.5729 | Recall: 0.8627 | AP: 0.6866
[6/48] Isolation Forest params: {'bootstrap': False, 'contamination': 'auto',
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6914 | Precision: 0.5748 | Recall: 0.8674 | AP: 0.6840
[7/48] Isolation Forest params: {'bootstrap': False, 'contamination': 'auto',
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6949 | Precision: 0.5839 | Recall: 0.8582 | AP: 0.7304
[8/48] Isolation Forest params: {'bootstrap': False, 'contamination': 'auto',
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6956 | Precision: 0.5847 | Recall: 0.8585 | AP: 0.7295
[9/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.01,
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6779 | Precision: 0.5672 | Recall: 0.8423 | AP: 0.6709
[10/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.01,
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6820 | Precision: 0.5745 | Recall: 0.8388 | AP: 0.6798
[11/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.01,
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6937 | Precision: 0.5821 | Recall: 0.8584 | AP: 0.7228
[12/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.01,
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6959 | Precision: 0.6066 | Recall: 0.8161 | AP: 0.7315
[13/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.01,
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6886 | Precision: 0.5729 | Recall: 0.8627 | AP: 0.6866
```

```
[14/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.01,
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6914 | Precision: 0.5748 | Recall: 0.8674 | AP: 0.6840
[15/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.01,
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6949 | Precision: 0.5839 | Recall: 0.8582 | AP: 0.7304
[16/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.01,
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6956 | Precision: 0.5847 | Recall: 0.8585 | AP: 0.7295
[17/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.03,
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6779 | Precision: 0.5672 | Recall: 0.8423 | AP: 0.6709
[18/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.03,
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6820 | Precision: 0.5745 | Recall: 0.8388 | AP: 0.6798
[19/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.03,
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6937 | Precision: 0.5821 | Recall: 0.8584 | AP: 0.7228
[20/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.03,
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6959 | Precision: 0.6066 | Recall: 0.8161 | AP: 0.7315
[21/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.03,
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6886 | Precision: 0.5729 | Recall: 0.8627 | AP: 0.6866
[22/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.03,
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6914 | Precision: 0.5748 | Recall: 0.8674 | AP: 0.6840
[23/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.03,
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6949 | Precision: 0.5839 | Recall: 0.8582 | AP: 0.7304
[24/48] Isolation Forest params: {'bootstrap': False, 'contamination': 0.03,
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6956 | Precision: 0.5847 | Recall: 0.8585 | AP: 0.7295
[25/48] Isolation Forest params: {'bootstrap': True, 'contamination': 'auto',
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6780 | Precision: 0.5769 | Recall: 0.8222 | AP: 0.6704
[26/48] Isolation Forest params: {'bootstrap': True, 'contamination': 'auto',
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6819 | Precision: 0.5843 | Recall: 0.8186 | AP: 0.6796
[27/48] Isolation Forest params: {'bootstrap': True, 'contamination': 'auto',
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6952 | Precision: 0.5869 | Recall: 0.8526 | AP: 0.7221
[28/48] Isolation Forest params: {'bootstrap': True, 'contamination': 'auto',
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6961 | Precision: 0.5881 | Recall: 0.8528 | AP: 0.7258
    New BEST Isolation Forest (F1: 0.6961)
[29/48] Isolation Forest params: {'bootstrap': True, 'contamination': 'auto',
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 300}
```

```
   -> F1: 0.6882 | Precision: 0.5853 | Recall: 0.8350 | AP: 0.6878
[30/48] Isolation Forest params: {'bootstrap': True, 'contamination': 'auto',
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6902 | Precision: 0.5806 | Recall: 0.8510 | AP: 0.6846
[31/48] Isolation Forest params: {'bootstrap': True, 'contamination': 'auto',
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6943 | Precision: 0.5726 | Recall: 0.8817 | AP: 0.7272
[32/48] Isolation Forest params: {'bootstrap': True, 'contamination': 'auto',
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6980 | Precision: 0.5821 | Recall: 0.8716 | AP: 0.7328
    New BEST Isolation Forest (F1: 0.6980)
[33/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.01,
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6780 | Precision: 0.5769 | Recall: 0.8222 | AP: 0.6704
[34/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.01,
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6819 | Precision: 0.5843 | Recall: 0.8186 | AP: 0.6796
[35/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.01,
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6952 | Precision: 0.5869 | Recall: 0.8526 | AP: 0.7221
[36/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.01,
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6961 | Precision: 0.5881 | Recall: 0.8528 | AP: 0.7258
[37/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.01,
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6882 | Precision: 0.5853 | Recall: 0.8350 | AP: 0.6878
[38/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.01,
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6902 | Precision: 0.5806 | Recall: 0.8510 | AP: 0.6846
[39/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.01,
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6943 | Precision: 0.5726 | Recall: 0.8817 | AP: 0.7272
[40/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.01,
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6980 | Precision: 0.5821 | Recall: 0.8716 | AP: 0.7328
[41/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.03,
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6780 | Precision: 0.5769 | Recall: 0.8222 | AP: 0.6704
[42/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.03,
'max_features': 0.8, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6819 | Precision: 0.5843 | Recall: 0.8186 | AP: 0.6796
[43/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.03,
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6952 | Precision: 0.5869 | Recall: 0.8526 | AP: 0.7221
[44/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.03,
'max_features': 0.8, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6961 | Precision: 0.5881 | Recall: 0.8528 | AP: 0.7258
[45/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.03,
```

```
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 300}
   -> F1: 0.6882 | Precision: 0.5853 | Recall: 0.8350 | AP: 0.6878
[46/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.03,
'max_features': 1.0, 'max_samples': 'auto', 'n_estimators': 500}
   -> F1: 0.6902 | Precision: 0.5806 | Recall: 0.8510 | AP: 0.6846
[47/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.03,
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 300}
   -> F1: 0.6943 | Precision: 0.5726 | Recall: 0.8817 | AP: 0.7272
[48/48] Isolation Forest params: {'bootstrap': True, 'contamination': 0.03,
'max_features': 1.0, 'max_samples': 0.8, 'n_estimators': 500}
   -> F1: 0.6980 | Precision: 0.5821 | Recall: 0.8716 | AP: 0.7328
```

	model	bootstrap	contamination	max_features	max_samples	\
0	Isolation Forest	True	auto	1.0	0.8	

	n_estimators	threshold	precision	recall	f1	roc_auc	\
0	500	-0.128215	0.582084	0.871601	0.698012	0.754499	

	avg_precision
0	0.732775

1.4.2 2. Local Outlier Factor in novelty mode

[ ]:
```
print("Training and tuning Local Outlier Factor...")

X_train_lof = X_train_scaled
if MAX_BENIGN_TRAIN_SAMPLES_LOF is not None and len(X_train_lof) >␣
 ↪MAX_BENIGN_TRAIN_SAMPLES_LOF:
    X_train_lof = pd.DataFrame(X_train_lof).sample(
        n=MAX_BENIGN_TRAIN_SAMPLES_LOF,
        random_state=SEED,
    ).to_numpy()

lof_params = {
    "n_neighbors": [15, 25, 35],
    "leaf_size": [20, 30],
    "metric": ["minkowski"],
    "p": [1, 2],
    "contamination": ["auto", 0.01, 0.03],
    "novelty": [True],
}

best_lof = None
best_lof_model = None
best_lof_scores_val = None

param_list = list(ParameterGrid(lof_params))
total_runs = len(param_list)
```

```
for i, params in enumerate(param_list, 1):
    print(f"[{i}/{total_runs}] LOF params: {params}")

    candidate = LocalOutlierFactor(**params)
    candidate.fit(X_train_lof)

    val_scores = -candidate.decision_function(X_val_scaled)
    threshold_info = tune_threshold(
        y_val,
        val_scores,
        objective=THRESHOLD_OBJECTIVE,
        min_recall=MIN_RECALL,
    )

    row = {
        "model": "Local Outlier Factor",
        **params,
        **threshold_info,
        **score_summary(y_val, val_scores),
    }

    print(
        f"   -> F1: {row['f1']:.4f} | Precision: {row['precision']:.4f} | "
        f"Recall: {row['recall']:.4f} | AP: {row['avg_precision']:.4f}"
    )

    if best_lof is None or (row["f1"], row["avg_precision"]) > (best_lof["f1"],␣
↪best_lof["avg_precision"]):
        best_lof = row
        best_lof_model = candidate
        best_lof_scores_val = val_scores
        print(f"***New BEST LOF (F1: {row['f1']:.4f})")

display(pd.DataFrame([best_lof]))
```

```
Training and tuning Local Outlier Factor…
[1/36] LOF params: {'contamination': 'auto', 'leaf_size': 20, 'metric':
'minkowski', 'n_neighbors': 15, 'novelty': True, 'p': 1}
   -> F1: 0.6783 | Precision: 0.5849 | Recall: 0.8073 | AP: 0.7553
***New BEST LOF (F1: 0.6783)
[2/36] LOF params: {'contamination': 'auto', 'leaf_size': 20, 'metric':
'minkowski', 'n_neighbors': 15, 'novelty': True, 'p': 2}
   -> F1: 0.6488 | Precision: 0.5013 | Recall: 0.9192 | AP: 0.7336
[3/36] LOF params: {'contamination': 'auto', 'leaf_size': 20, 'metric':
'minkowski', 'n_neighbors': 25, 'novelty': True, 'p': 1}
   -> F1: 0.6794 | Precision: 0.5846 | Recall: 0.8110 | AP: 0.7568
***New BEST LOF (F1: 0.6794)
```

```
[4/36] LOF params: {'contamination': 'auto', 'leaf_size': 20, 'metric':
'minkowski', 'n_neighbors': 25, 'novelty': True, 'p': 2}
   -> F1: 0.6506 | Precision: 0.5010 | Recall: 0.9273 | AP: 0.7334
[5/36] LOF params: {'contamination': 'auto', 'leaf_size': 20, 'metric':
'minkowski', 'n_neighbors': 35, 'novelty': True, 'p': 1}
   -> F1: 0.6808 | Precision: 0.6253 | Recall: 0.7471 | AP: 0.7542
***New BEST LOF (F1: 0.6808)
[6/36] LOF params: {'contamination': 'auto', 'leaf_size': 20, 'metric':
'minkowski', 'n_neighbors': 35, 'novelty': True, 'p': 2}
   -> F1: 0.6502 | Precision: 0.4994 | Recall: 0.9313 | AP: 0.7291
[7/36] LOF params: {'contamination': 'auto', 'leaf_size': 30, 'metric':
'minkowski', 'n_neighbors': 15, 'novelty': True, 'p': 1}
   -> F1: 0.6783 | Precision: 0.5849 | Recall: 0.8073 | AP: 0.7553
[8/36] LOF params: {'contamination': 'auto', 'leaf_size': 30, 'metric':
'minkowski', 'n_neighbors': 15, 'novelty': True, 'p': 2}
   -> F1: 0.6488 | Precision: 0.5013 | Recall: 0.9192 | AP: 0.7336
[9/36] LOF params: {'contamination': 'auto', 'leaf_size': 30, 'metric':
'minkowski', 'n_neighbors': 25, 'novelty': True, 'p': 1}
   -> F1: 0.6794 | Precision: 0.5846 | Recall: 0.8110 | AP: 0.7568
[10/36] LOF params: {'contamination': 'auto', 'leaf_size': 30, 'metric':
'minkowski', 'n_neighbors': 25, 'novelty': True, 'p': 2}
   -> F1: 0.6506 | Precision: 0.5010 | Recall: 0.9273 | AP: 0.7334
[11/36] LOF params: {'contamination': 'auto', 'leaf_size': 30, 'metric':
'minkowski', 'n_neighbors': 35, 'novelty': True, 'p': 1}
```

1.4.3 3. Denoising deep autoencoder

[21]:
```python
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

X_train_ae, X_holdout_ae = train_test_split(
    X_train_scaled,
    test_size=0.10,
    random_state=SEED,
)

class DenoisingAutoencoder(nn.Module):
    def __init__(self, input_dim):
        super().__init__()
        self.encoder = nn.Sequential(
            nn.Linear(input_dim, 128),
            nn.BatchNorm1d(128),
            nn.LeakyReLU(0.1),
            nn.Dropout(0.10),

            nn.Linear(128, 64),
            nn.BatchNorm1d(64),
```

```
            nn.LeakyReLU(0.1),
            nn.Dropout(0.10),

            nn.Linear(64, 32),
            nn.LeakyReLU(0.1),

            nn.Linear(32, 12)
        )

        self.decoder = nn.Sequential(
            nn.Linear(12, 32),
            nn.LeakyReLU(0.1),

            nn.Linear(32, 64),
            nn.BatchNorm1d(64),
            nn.LeakyReLU(0.1),

            nn.Linear(64, 128),
            nn.BatchNorm1d(128),
            nn.LeakyReLU(0.1),

            nn.Linear(128, input_dim)
        )

    def forward(self, x):
        z = self.encoder(x)
        out = self.decoder(z)
        return out

input_dim = X_train_scaled.shape[1]
ae_model = DenoisingAutoencoder(input_dim).to(device)

criterion = nn.SmoothL1Loss()
optimizer = optim.Adam(
    ae_model.parameters(),
    lr=AE_LEARNING_RATE,
    weight_decay=AE_WEIGHT_DECAY,
)

scheduler = optim.lr_scheduler.ReduceLROnPlateau(
    optimizer,
    mode="min",
    factor=0.5,
    patience=4,
)

train_tensor = torch.tensor(X_train_ae, dtype=torch.float32)
```

```
holdout_tensor = torch.tensor(X_holdout_ae, dtype=torch.float32)

train_loader = DataLoader(
    TensorDataset(train_tensor),
    batch_size=BATCH_SIZE,
    shuffle=True,
    drop_last=False,
)

best_state = None
best_holdout_loss = np.inf
epochs_without_improvement = 0
history = []

print("Training denoising autoencoder...")

for epoch in range(AE_EPOCHS):
    ae_model.train()
    train_losses = []

    for batch in train_loader:
        clean_inputs = batch[0].to(device)
        noisy_inputs = clean_inputs + AE_NOISE_STD * torch.
 ↪randn_like(clean_inputs)

        optimizer.zero_grad()
        outputs = ae_model(noisy_inputs)
        loss = criterion(outputs, clean_inputs)
        loss.backward()
        optimizer.step()

        train_losses.append(loss.item())

    ae_model.eval()
    with torch.no_grad():
        holdout_inputs = holdout_tensor.to(device)
        holdout_outputs = ae_model(holdout_inputs)
        holdout_loss = criterion(holdout_outputs, holdout_inputs).item()

    train_loss = float(np.mean(train_losses))
    scheduler.step(holdout_loss)

    history.append({
        "epoch": epoch + 1,
        "train_loss": train_loss,
        "holdout_loss": holdout_loss,
    })
```

```
    if holdout_loss < best_holdout_loss:
        best_holdout_loss = holdout_loss
        best_state = {k: v.detach().cpu().clone() for k, v in ae_model.
 ↪state_dict().items()}
        epochs_without_improvement = 0
        print(f"[Epoch {epoch + 1}/{AE_EPOCHS}] ***New BEST holdout loss:␣
 ↪{holdout_loss:.6f}")
    else:
        epochs_without_improvement += 1

    if (epoch + 1) % 10 == 0 or epoch == 0:
        print(
            f"[Epoch {epoch + 1}/{AE_EPOCHS}] "
            f"train_loss: {train_loss:.6f} | holdout_loss: {holdout_loss:.6f}"
        )

    if epochs_without_improvement >= AE_PATIENCE:
        print(f"Early stopping at epoch {epoch + 1}.")
        break

if best_state is not None:
    ae_model.load_state_dict(best_state)

history_df = pd.DataFrame(history)
display(history_df.tail())

plt.figure(figsize=(10, 5))
plt.plot(history_df["epoch"], history_df["train_loss"], label="Train Loss")
plt.plot(history_df["epoch"], history_df["holdout_loss"], label="Holdout Loss")
plt.title("Denoising Autoencoder Training Curve")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.show()

# validation scoring
ae_model.eval()
with torch.no_grad():
    val_tensor = torch.tensor(X_val_scaled, dtype=torch.float32).to(device)
    val_reconstructions = ae_model(val_tensor)

    # average squared reconstruction error
    val_recon_error = torch.mean((val_tensor - val_reconstructions) ** 2,␣
 ↪dim=1).cpu().numpy()

    # max absolute error sometimes sharpens anomaly ranking
```

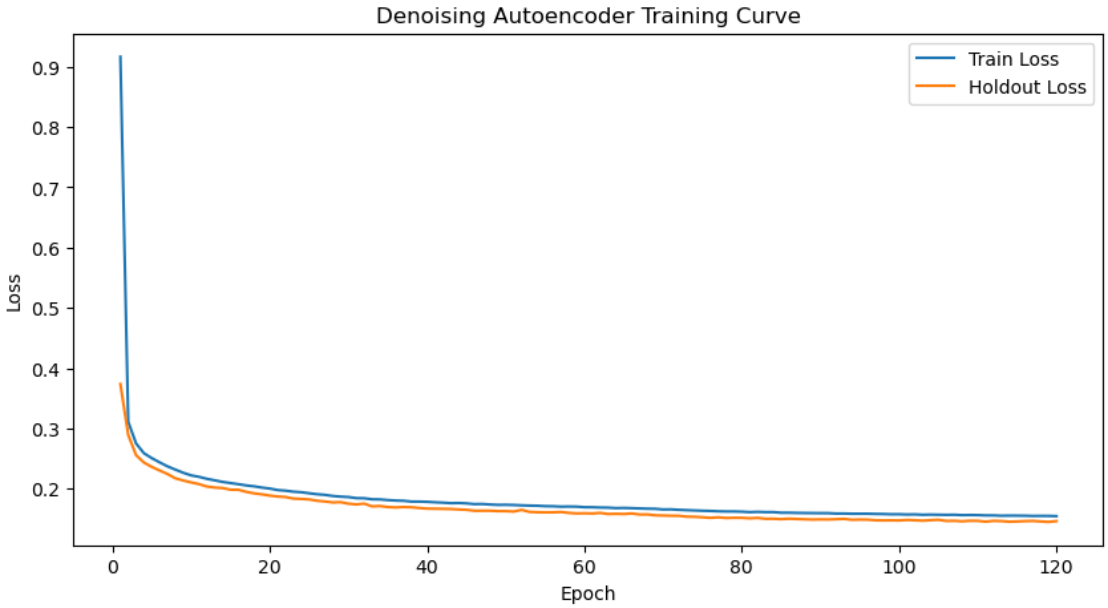
```
    val_max_abs_error = torch.max(torch.abs(val_tensor - val_reconstructions),␣
 ↪dim=1).values.cpu().numpy()

# hybrid reconstruction score
val_scores_ae = 0.7 * robust_zscore(val_recon_error) + 0.3 *␣
 ↪robust_zscore(val_max_abs_error)

threshold_info_ae = tune_threshold(
    y_val,
    val_scores_ae,
    objective=THRESHOLD_OBJECTIVE,
    min_recall=MIN_RECALL,
)

ae_val_row = {
    "model": "Denoising Deep Autoencoder",
    "threshold": threshold_info_ae["threshold"],
    "precision": threshold_info_ae["precision"],
    "recall": threshold_info_ae["recall"],
    "f1": threshold_info_ae["f1"],
    **score_summary(y_val, val_scores_ae),
}
display(pd.DataFrame([ae_val_row]))
```

```
Using device: cpu
Training denoising autoencoder…
[Epoch 1/120]   New BEST holdout loss: 0.373948
[Epoch 1/120] train_loss: 0.916523 | holdout_loss: 0.373948
[Epoch 2/120]   New BEST holdout loss: 0.289334
[Epoch 3/120]   New BEST holdout loss: 0.255895
[Epoch 4/120]   New BEST holdout loss: 0.243650
[Epoch 5/120]   New BEST holdout loss: 0.236429
[Epoch 6/120]   New BEST holdout loss: 0.230486
[Epoch 7/120]   New BEST holdout loss: 0.224633
[Epoch 8/120]   New BEST holdout loss: 0.217543
[Epoch 9/120]   New BEST holdout loss: 0.213948
[Epoch 10/120]   New BEST holdout loss: 0.210764
[Epoch 10/120] train_loss: 0.222425 | holdout_loss: 0.210764
[Epoch 11/120]   New BEST holdout loss: 0.207942
[Epoch 12/120]   New BEST holdout loss: 0.203944
[Epoch 13/120]   New BEST holdout loss: 0.202227
[Epoch 14/120]   New BEST holdout loss: 0.201375
[Epoch 15/120]   New BEST holdout loss: 0.198730
[Epoch 16/120]   New BEST holdout loss: 0.198579
[Epoch 17/120]   New BEST holdout loss: 0.195090
[Epoch 18/120]   New BEST holdout loss: 0.192689
[Epoch 19/120]   New BEST holdout loss: 0.190967
```

```
[Epoch 20/120]   New BEST holdout loss: 0.188884
[Epoch 20/120] train_loss: 0.200590 | holdout_loss: 0.188884
[Epoch 21/120]   New BEST holdout loss: 0.187385
[Epoch 22/120]   New BEST holdout loss: 0.186553
[Epoch 23/120]   New BEST holdout loss: 0.183935
[Epoch 24/120]   New BEST holdout loss: 0.183381
[Epoch 25/120]   New BEST holdout loss: 0.182555
[Epoch 26/120]   New BEST holdout loss: 0.180358
[Epoch 27/120]   New BEST holdout loss: 0.179156
[Epoch 28/120]   New BEST holdout loss: 0.177599
[Epoch 30/120]   New BEST holdout loss: 0.175440
[Epoch 30/120] train_loss: 0.186500 | holdout_loss: 0.175440
[Epoch 31/120]   New BEST holdout loss: 0.174280
[Epoch 33/120]   New BEST holdout loss: 0.171083
[Epoch 35/120]   New BEST holdout loss: 0.169949
[Epoch 36/120]   New BEST holdout loss: 0.169329
[Epoch 39/120]   New BEST holdout loss: 0.168333
[Epoch 40/120]   New BEST holdout loss: 0.167305
[Epoch 40/120] train_loss: 0.178579 | holdout_loss: 0.167305
[Epoch 41/120]   New BEST holdout loss: 0.167172
[Epoch 42/120]   New BEST holdout loss: 0.166906
[Epoch 43/120]   New BEST holdout loss: 0.166722
[Epoch 44/120]   New BEST holdout loss: 0.165966
[Epoch 45/120]   New BEST holdout loss: 0.165377
[Epoch 46/120]   New BEST holdout loss: 0.163750
[Epoch 49/120]   New BEST holdout loss: 0.163145
[Epoch 50/120] train_loss: 0.173696 | holdout_loss: 0.163159
[Epoch 51/120]   New BEST holdout loss: 0.162462
[Epoch 53/120]   New BEST holdout loss: 0.161921
[Epoch 54/120]   New BEST holdout loss: 0.161342
[Epoch 55/120]   New BEST holdout loss: 0.161122
[Epoch 58/120]   New BEST holdout loss: 0.160235
[Epoch 59/120]   New BEST holdout loss: 0.159395
[Epoch 60/120] train_loss: 0.169681 | holdout_loss: 0.159609
[Epoch 61/120]   New BEST holdout loss: 0.159303
[Epoch 63/120]   New BEST holdout loss: 0.158364
[Epoch 67/120]   New BEST holdout loss: 0.157570
[Epoch 69/120]   New BEST holdout loss: 0.156206
[Epoch 70/120]   New BEST holdout loss: 0.155889
[Epoch 70/120] train_loss: 0.165940 | holdout_loss: 0.155889
[Epoch 71/120]   New BEST holdout loss: 0.155504
[Epoch 72/120]   New BEST holdout loss: 0.155403
[Epoch 73/120]   New BEST holdout loss: 0.153973
[Epoch 74/120]   New BEST holdout loss: 0.153762
[Epoch 75/120]   New BEST holdout loss: 0.152981
[Epoch 76/120]   New BEST holdout loss: 0.152047
[Epoch 78/120]   New BEST holdout loss: 0.151789
[Epoch 80/120] train_loss: 0.162259 | holdout_loss: 0.152052
```

```
[Epoch 81/120]   New BEST holdout loss: 0.151244
[Epoch 83/120]   New BEST holdout loss: 0.150441
[Epoch 85/120]   New BEST holdout loss: 0.149820
[Epoch 88/120]   New BEST holdout loss: 0.149633
[Epoch 89/120]   New BEST holdout loss: 0.149189
[Epoch 90/120] train_loss: 0.159833 | holdout_loss: 0.149467
[Epoch 94/120]   New BEST holdout loss: 0.148815
[Epoch 97/120]   New BEST holdout loss: 0.147981
[Epoch 98/120]   New BEST holdout loss: 0.147735
[Epoch 100/120]   New BEST holdout loss: 0.147666
[Epoch 100/120] train_loss: 0.158001 | holdout_loss: 0.147666
[Epoch 103/120]   New BEST holdout loss: 0.147316
[Epoch 106/120]   New BEST holdout loss: 0.146892
[Epoch 108/120]   New BEST holdout loss: 0.146270
[Epoch 110/120] train_loss: 0.156591 | holdout_loss: 0.147015
[Epoch 111/120]   New BEST holdout loss: 0.145706
[Epoch 114/120]   New BEST holdout loss: 0.145601
[Epoch 119/120]   New BEST holdout loss: 0.145258
[Epoch 120/120] train_loss: 0.154876 | holdout_loss: 0.146543
```

	epoch	train_loss	holdout_loss
115	116	0.155554	0.146549
116	117	0.155182	0.146934
117	118	0.155267	0.146007
118	119	0.155222	0.145258
119	120	0.154876	0.146543




```
                model  threshold  precision    recall        f1  \
```

```
0  Denoising Deep Autoencoder  -0.223521   0.648685  0.819839  0.724288

    roc_auc  avg_precision
0  0.770946       0.711621
```

1.4.4 Validation comparison

[ ]:
```
threshold_if = best_if["threshold"]
val_preds_if = (best_if_scores_val > threshold_if).astype(int)
res_if = evaluate_model(y_val, val_preds_if, "Isolation Forest", "validation")
plot_score_distributions(y_val, best_if_scores_val, "Isolation Forest")
plot_confusion(y_val, val_preds_if, "Isolation Forest - Validation")
summarize_business_view(y_val, val_preds_if)

threshold_lof = best_lof["threshold"]
val_preds_lof = (best_lof_scores_val > threshold_lof).astype(int)
res_lof = evaluate_model(y_val, val_preds_lof, "Local Outlier Factor",␣
 ↪"validation")
plot_score_distributions(y_val, best_lof_scores_val, "Local Outlier Factor")
plot_confusion(y_val, val_preds_lof, "Local Outlier Factor - Validation")
summarize_business_view(y_val, val_preds_lof)

threshold_ae = ae_val_row["threshold"]
val_preds_ae = (val_scores_ae > threshold_ae).astype(int)
res_ae = evaluate_model(y_val, val_preds_ae, "Denoising Deep Autoencoder",␣
 ↪"validation")
plot_score_distributions(y_val, val_scores_ae, "Denoising Deep Autoencoder")
plot_confusion(y_val, val_preds_ae, "Denoising Deep Autoencoder - Validation")
summarize_business_view(y_val, val_preds_ae)
```

1.4.5 Summary

[46]:
```
validation_summary = pd.DataFrame([res_if, res_lof, res_ae]).sort_values(
    by=["f1", "recall", "precision"],
    ascending=False,
).reset_index(drop=True)

display(validation_summary)
```

```
                        model       split        f1  precision    recall
0  Denoising Deep Autoencoder  validation  0.724288   0.648685  0.819839
1            Isolation Forest  validation  0.698012   0.582084  0.871601
2        Local Outlier Factor  validation  0.680830   0.625337  0.747130
```